ISB 140/DS 110−INTRODUCTION TO STATISTICS

LECTURE NOTES

UNIT III: SKEWNESS, KURTOSIS, AND CURVE FITTING

1. SKEWNESS

- ✓ The general shape of the distribution of data can be easily seen if the distributions represented by a histogram or a stem and leaf display are illustrated using a frequency curve.
- ✓ Some of these distributions are symmetrical while others are skewed. Besides, some are more peaked while others are flattered.
- ✓ Thus, when determining the general shape of distribution of data, its important to consider two aspects of measures of distribution such as **Skewness** and **kurtosis.**
- ✓ If there are large extreme values in the data the mean is pulled to the right or left and we say that the distribution exhibits skewness or kurtosis.
- ✓ **Skewness** literally means lack of symmetry*;* thus, skewness is the measure of the degree of asymmetry of the frequency distribution.
- ✓ We study skewness to have an idea about the shape of the curve which we can draw with the help of the given data.
- ✓ A distribution is said to be skewed if and only if:

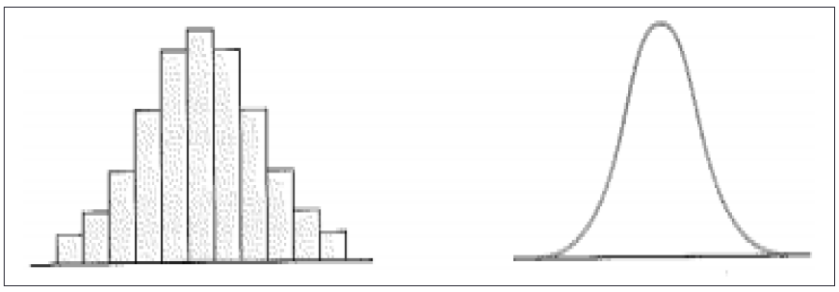
  (i) Mean, median and mode fall at different points.

  $$i.e., Mean \neq Median \neq Mode$$

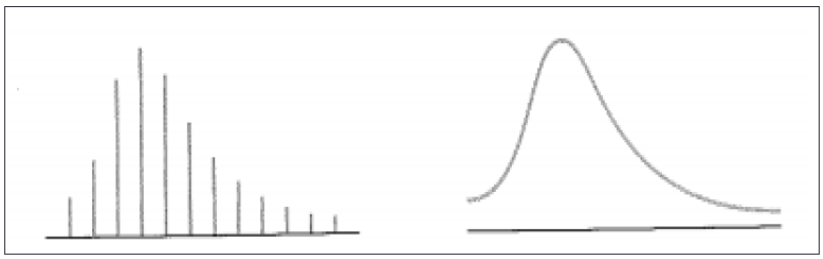
  (ii) Quartiles are not equidistant from median, and

  (iii) The curve drawn with the help of the given data is not symmetrical but stretched more to one side than to the other.
- ✓ There are three types of Skewness of data distribution, these are zero skewed, right-skewed (positively skewed) and left skewed (negatively skewed).

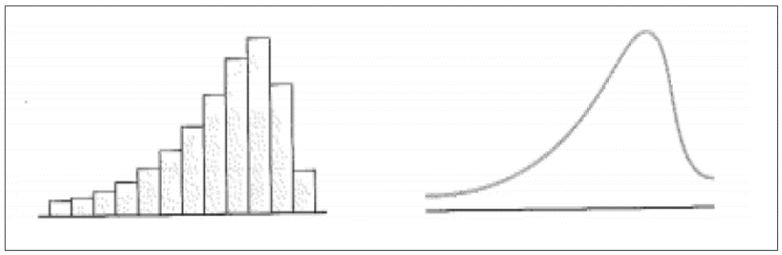
  (a) **A symmetrical or bell-shaped or normal distribution**: is the distribution where the skew is neither pulled to the left nor to the right. In brief it is a distribution where the skew is zero. It is diagrammatically shown below.



(b) A right-skewed or positively skewed distribution: is a distribution that stretches asymmetrically to the right than it does to the left as indicated in the diagram below

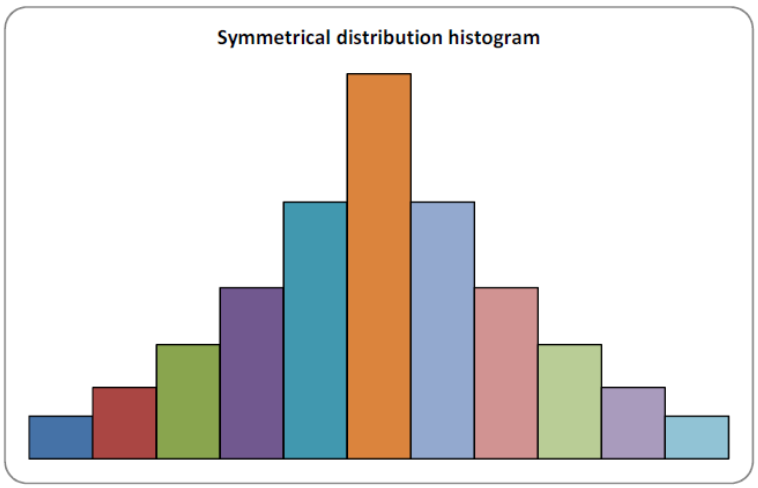


(c) A left-skewed or negatively skewed distribution: is a distribution that stretches asymmetrically to the left than it does to the right as indicated in the diagram below.



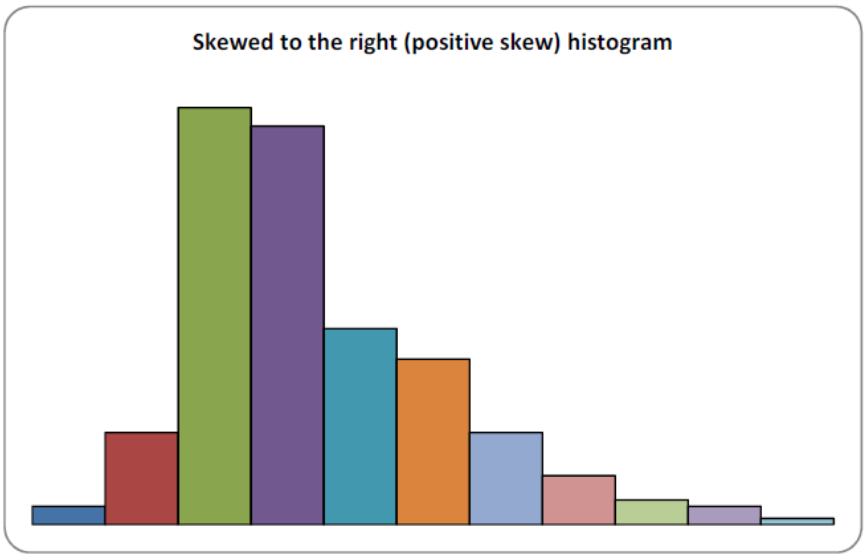
✓ Thus, there are mathematical ways of expressing the degree of skewness of a distibution of data by comparing the mean, mode and the median as expressed below:

1. For a **symmetrical distribution** or **normal distribution,** the mean, median and mode will be about the same or coincide, that is, $\text{mean} = \text{mode} = \text{median}$.



2. For a distribution that is **skewed to the right (or positively skewed distribution),** the tail of the distribution is pulled to the right or in the positive direction, and so the right tail is longer than the left tail. Thus, in a positively skewed distribution (or right-skewed), the mode will be less than the median and the median will be less

than the mean, that is, $mode < median < mean \Leftrightarrow mean > median > mode$.



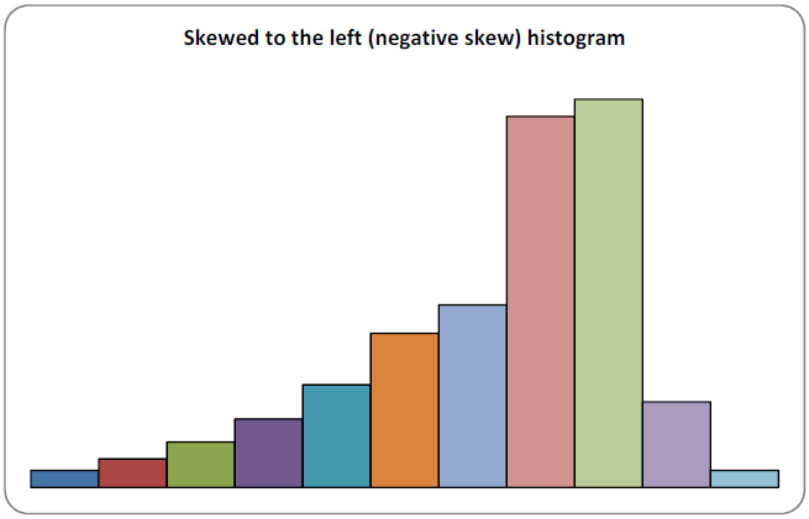
NOTE

(i) In a right-skewed (or a positively skewed) distribution, there is a long tail at the positive (or right) end of the distribution.

(ii) In a right-skewed, most of the data is concentrated to the left of the histogram.

(iii) In a right-skewed, the relative position of the mean is such that the mean is higher than the median which is higher than the mode.

3. For a distribution that is **skewed to the left (negatively skewed)**, the tail of the distribution is pulled to the left or in a negative direction , and so the left tail is longer than the right tail. Thus, in a negatively skewed distribution or a left-skewed distribution, the mean is the smallest, followed by the median, while the mode is the largest, that is, mean $<$ median $<$ mode $\Leftrightarrow$ mode $>$ median $>$ mean.



NOTE

(i) In a left-skewed or a negatively skewed distribution there is a long tail at the

negative (or left) end of the distribution.

(ii) In a left-skewed most of the data is clustered to the right of the histogram

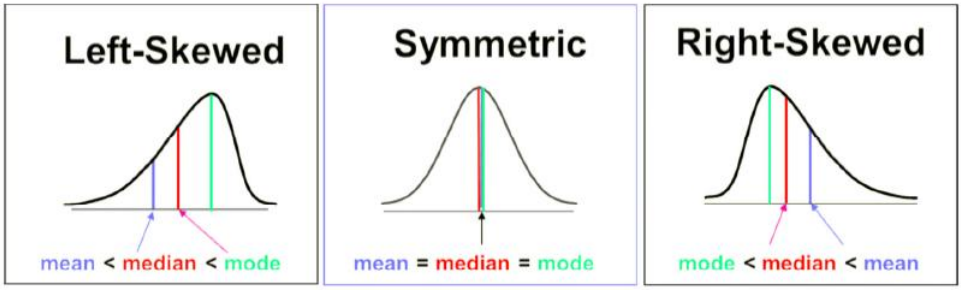
(iii) In a left-skewed the relative position of the measures of central tendency are reversed, that is, the mean is lower than the median which is lower than the mode.

SUMMARY

1. As a general rule the difference between the median and the mode is about twice the difference between the mean and the median.
2. If the data are skewed to the left there are some outliers on the left (small values).
3. If the data are skewed to the right then there are some large outliers.
5. When the distributions are skewed, the median generally lies between the mode and the mean, and the following relationship is satisfied.

$$mean - mode = 3(mean - median)$$

6. A comparison of the mean and median can reveal information about skewness.
7. Data can be identified as skewed to the left, symmetrical or skewed to the right.
8. Data skewed to the left will have the mean and median to the left of the mode.
9. Data skewed to the right will have the mean and median to the right of the mode.



1.1. MEASURES OF SKEWNESS OF A DISTRIBUTION

- ✓ While measures of central tendency have to do with centrality of the data and measures of variability measure the spread of the data, measures of skewness show the extent of lopsidedness or the degree of asymmetry in the data.
- ✓ There are Various absolute measures of skewness which include:

  (i) $S_k = Mean - Median$

  (ii) $S_k = Mean - Mode$

  (iii) $S_k = (Q_3 - Q_2) - (Q_2 - Q_1)$
- ✓ As in dispersion, for comparing two series, we do not calculate these absolute measures of skewness but we calculate the relative measures of skewness called the

coefficients of skewness which are pure numbers independent of units of measurement.

- ✓ coefficients of skewness include among others, Pearson's coefficient of skewness (PS_k), Quartile coefficient of skewness (QS_k) and the coefficient of skewness based on moment (S_k).

1.2. PEARSON'S COEFFICIENT OF SKEWNESS

- ✓ Pearson's coefficient of skewness also called the Karl pearson's coefficient of skewness, denoted as PS_k, is a measure of the degree of departure from symmetry which is based on the difference between the mean and the median.
- ✓ It is calculated using the formula as follows:

$$Pearson's\ coefficient\ of\ skewness\ (PS_k) = \frac{mean - mode}{standard\ deviation}$$
$$= \frac{\bar{x} - mode}{S}$$

- ✓ **NOTE**

  1. If $mean > mode$, then the skew is posive.
  2. If $mean < mode$, then the skew is negative
  3. If $mean = mode$, then the skew is zero and the distribution is symmetrical.

- ✓ Alternatively, if mode is ill-defined, then using the relation

  $mode = 3median - 2mean$ for a moderately asymmetrical distribution, we get

$$Pearson's\ coefficient\ of\ skewness(PS_k) = \frac{3\ (mean - median)}{standard\ deviation}$$
$$= \frac{3\ (\bar{x} - Q_2)}{s}$$

- ✓ **NOTE**

  1. The pearson's coefficient of skewness is also known as skewness
  2. Skewness can take any value between -3 and 3 ($limits\ of\ PS_k$).

1.3. QUARTILE COEFFICIENT OF SKEWNESS

- ✓ Quartile coefficient of skewness also known as Bowley's coefficient of skewness is another relative measure of skewness which is defined interms of the quartiles such as the lower quartile (Q_1), median (Q_2) and upper quartile (Q_3).
- ✓ It is calculated using the formula below:

$$Quartile\ coefficient\ of\ skewness\ (QS_k) = \frac{(Q_3 - Q_2) - (Q_2 - Q_1)}{Q_3 - Q_1}$$
$$= \frac{Q_3 + Q_1 - 2Q_2}{Q_3 - Q_1}$$

- ✓ Bowley's coefficient of skewness is also known as Quartile coefficient of skewness is very useful in the following:

  (i) When the mode is ill-defined (unclear) and extreme observation are present in the data.

  (ii) When the distribution has open end classes or unequal class intervals.

- ✓ Thus, in these situations Pearson's coefficient of skewness cannot be used.

REMARK

We have observed the following about the quartile coefficient of skewness (qsk):

1. For a symmetrical distribution median is equidistant from the upper and lower quartiles, that is, $qsk = 0$, if $Q_3 - Q_2 = Q_2 - Q_1$.

2. Quartile coefficient of skewness (qsk) is positive if

$$Q_3 - Q_2 > Q_2 - Q_1 \Longrightarrow Q_3 + Q_1 > 2Q_2$$

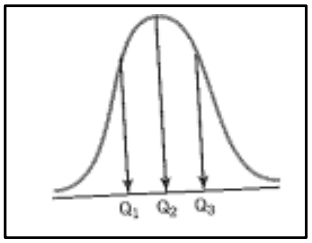
3 Quartile coefficient of skewness (qsk) is negative if

$$Q_3 - Q_2 < Q_2 - Q_1 \Longrightarrow Q_3 + Q_1 < 2Q_2$$

4. Bowley's coefficient of skewness (quartile coefficient of skewness) ranges from -1 to 1 (Limits of QS_k).

SUMMARY

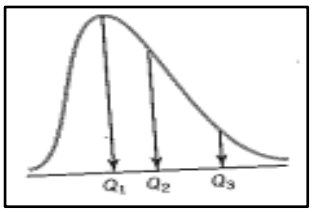
Symetrical Distribution



$Q_3 - Q_2 = Q_2 - Q_1$

Quartile skewness $= 0$

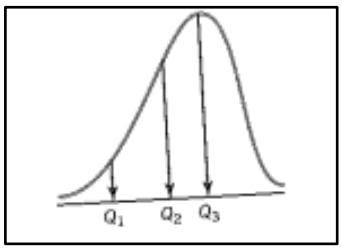
Positively Skewed Distribution



$Q_3 - Q_2 > Q_2 - Q_1$

Quartile skewness > 0

Negatively Skewed Distribution



$Q_3 - Q_2 < Q_2 - Q_1$

Quartile skewness < 0

NOTE

The two coefficients (PS_k and QS_k) are interpreted as follows:

(1) If a coefficient of skewness is negative, that is, $PS_k, QS_k < 0$, then it means or signifies that data is negatively skewed or left skewed.

(2) If a coefficient of skewness is equal to zero, that is, $PS_k, QS_k = 0$, then it shows that there is no skew, meaning that the distribution is symmetrical.

(3) If A coefficient of skewness is positive, that is, $PS_k, QS_k > 0$, then it signifies or shows that data is positively skewed or right skewed.

(4) The greater the value of PS_k or QS_k, the more skewed the distribution is.

(5) If a coefficient of skewness is close to zero, then it indicates that there is moderate skewness.

(6) It should be clearly understood that the values of the coefficients of skewness obtained by Bowley's formula and Pearson's formula are not comparable, although in each case, $PS_k, QS_k = 0$, implies the absence of skewness, *i.e.*, the distribution is symmetrical. It may even happen that one of them gives positive skewness while the other gives negative skewness.

EXAMPLES

1. Electric fuses normally rated for 30 Ampheres (30A), are tested by passing a gradually increasing electric current through them and recording the current , x ampheres, at which they blow. the result of this test on sample of 125 fuses are shown in the following table.

Current (x A)	Number of fuses
$25 \leq x < 28$	6
$28 \leq x < 29$	12
$29 \leq x < 30$	27
$30 \leq x < 31$	30
$31 \leq x < 32$	18
$32 \leq x < 33$	14
$33 \leq x < 34$	9
$34 \leq x < 35$	4
$35 \leq x < 40$	5

(a) Draw the histogram to represent these data

(b) For this sample calculate

(i) The median and the mode current,

(ii) The mean current,

(iii) The standard deviation of current

(iv) The lower and upper quartiles

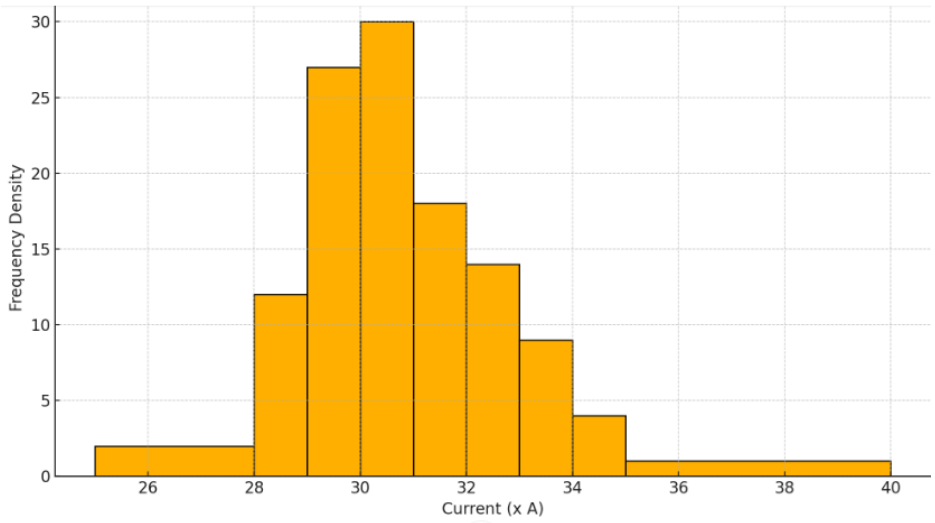
(c) (i) Compute the value of the two measures of skewness for the above data

(ii) Explain briefly on how this skewness is apparent in the shape of your histogram.

SOLUTION

(a)

Current (x A)	Frequency	Interval width	$Frequency\ density = \frac{frequency}{interval\ width}$
$25 \leq x < 28$	6	3	2
$28 \leq x < 29$	12	1	12
$29 \leq x < 30$	27	1	27
$30 \leq x < 31$	30	1	30
$31 \leq x < 32$	18	1	18
$32 \leq x < 33$	14	1	14
$33 \leq x < 34$	9	1	9
$34 \leq x < 35$	4	1	4
$35 \leq x < 40$	5	5	1



(b)

Current (x A)	Frequency	cf	Midpoint (x)	fx	fx^2

$25 \leq x < 28$	6	6	26.5	159	4213.5
$28 \leq x < 29$	12	18	28.5	342	9747
$29 \leq x < 30$	27	45	29.5	796.5	23496.75
$30 \leq x < 31$	30	75	30.5	915	27907.5
$31 \leq x < 32$	18	93	31.5	567	17860.5
$32 \leq x < 33$	14	107	32.5	455	14787.5
$33 \leq x < 34$	9	116	33.5	301.5	10100.25
$34 \leq x < 35$	4	120	34.5	138	4761
$35 \leq x < 40$	5	125	37.5	187.5	7031.25
Total	$\sum f = 125$			$\sum fx = 3861.5$	$\sum fx^2 = 119905.25$

(i) The median current

$Q_2\ position = (N/2)^{th} = (125/2)^{th} = 62.5\ th\ value$

$Q_2\ class = 30 - 31,$

$$\therefore \text{Median } (Q_2) = L_{Q_2} + \left(\frac{\frac{N}{2} - CF}{f_{Q_2}}\right) i$$

$$= 30 + \left(\frac{62.5 - 45}{30}\right) \times 1 = 30 + \left(\frac{17.5}{30}\right) \times 1 = 30.58\ A$$

$modal\ class$: $30 - 31$ (has 30 the highest frequency number)

$\Delta_1 = 30 - 27 = 3$, $\Delta_2 = 30 - 18 = 12$ and $i = 1$

$$\therefore Mode = L_{mo} + \left(\frac{\Delta_1}{\Delta_1 + \Delta_2}\right) i$$

$$= 30 + \left(\frac{3}{3 + 12}\right) 1 = 30.2\ A$$

(ii) The mean current,

$$mean\ (\bar{x}) = \frac{\sum fx}{\sum f} = \frac{3861.5}{125} = 30.89A$$

(iii) standard deviation $(S) = \sqrt{\frac{\sum fx^2}{\sum f} - (\bar{x})^2}$

$$= \sqrt{\frac{119905.25}{125} - (30.892)^2} = \sqrt{4.926\ldots} = 2.22A$$

(vi) $Q_1\ position = (N/4)^{th} = (125/4)^{th} = 31.25\ th\ value$

$Q_1\ class = 29 - 30,$

$$\therefore \text{Lower quartile}(Q_1) = L_{Q_1} + \left(\frac{\frac{N}{4} - CF}{f_{Q_1}}\right) i$$

$$= 29 + \left(\frac{31.25 - 18}{27}\right) \times 1 = 29 + \left(\frac{13.25}{27}\right) 1 = 29.49\ A$$

$$Q_3\ position = (3N/4)^{th} = (3 \times 125/4)^{th} = 93.75\ th\ value$$

$$Q_3\ class = 32 - 33,$$

$$\therefore \text{upper quartile } (Q_3) = L_{Q_3} + \left(\frac{\frac{3N}{4} - CF}{f_{Q_3}}\right) i$$

$$= 32 + \left(\frac{93.75 - 93}{14}\right) 1 = 32 + \left(\frac{0.75}{14}\right) 1 = 32.05\ A$$

(c) (i) The following are the computations of the two coefficients of skewness:

$$Pearson's\ coefficient\ of\ skewness\ (PS_k) = \frac{mean - mode}{standard\ deviation}$$

$$= \frac{\bar{x} - mode}{S}$$

$$= \frac{30.89A - 30.2\ A}{2.22A} = 0.31$$

$$\Leftrightarrow Pearson's\ coefficient\ of\ skewness\ (PS_k) = \frac{3\ (mean - median)}{standard\ deviation}$$

$$= \frac{3\ (\bar{x} - Q_2)}{s}$$

$$= \frac{3\ (30.89A - 30.58\ A)}{2.22A} = 0.42$$

$$Quartile\ coefficient\ of\ skewness\ (QS_k) = \frac{(Q_3 - Q_2) - (Q_2 - Q_1)}{Q_3 - Q_1}$$

$$= \frac{Q_3 + Q_1 - 2Q_2}{Q_3 - Q_1}$$

$$= \frac{32.05\ A + 29.49\ A - 2 \times 30.58\ A}{32.05\ A - 29.49\ A}$$

$$= \frac{61.54 - 61.16}{2.56} = 0.15\ (2dp)$$

(ii) Both skewness are positive, that is, $PS_k,\ QS_k > 0$ and that $mode < median < mean$, then the distribution is positively skewed which is apparent in the shape of the histogram where the right tail is longer than the left tail indicating a positively skewed distribution.

2. 31 KMU 4th year students from the Department of ICT tried to estimate the length of

a line. The line was actually 60mm. the following were their results in milimetres. Find the median and the quartiles of this distribution and use the quartiles to estimate the skewness. Draw the histogram with equal class intervals $20 \leq l \leq 30,\ 30 \leq l < 40, \ldots$

61	70	46	44	26	23	30	83	52	44	38
37	49	59	58	63	31	29	37	48	76	61
46	31	38	41	49	52	56	75	61		

SOLUTION

First arrange the result in order

23	26	29	30	31	31	37	37	38	38	41	44	44	46	46
48	49	49	52	52	56	58	59	61	61	61	63	70	75	76	83

There are 31 results, implying $n = 31$ is odd. Since n is odd, then the required quartiles are obtained as follows:

$$Q_2\ position = (n + 1/2)^{th} = (31 + 1/2)^{th} = 16th\ value.$$

$$\therefore median\ (Q_2) = 48$$

$$Q_1\ position = (n + 1/4)^{th} = (31 + 1/4)^{th} = 8th\ value.$$

$$\therefore Lower\ quartile\ (Q_1) = 37$$

$$Q_3\ position = 3(n + 1/4)^{th} = 3(31 + 1/4)^{th} = 24th\ value.$$

$$\therefore Upper\ quartile\ (Q_3) = 61$$

$$Q_3 - Q_2 = 61 - 48 = 13$$

$$Q_2 - Q_1 = 48 - 37 = 11$$

Since $Q_3 - Q_2 > Q_2 - Q_1$, the distribution is positively skewed.

Alternatively,

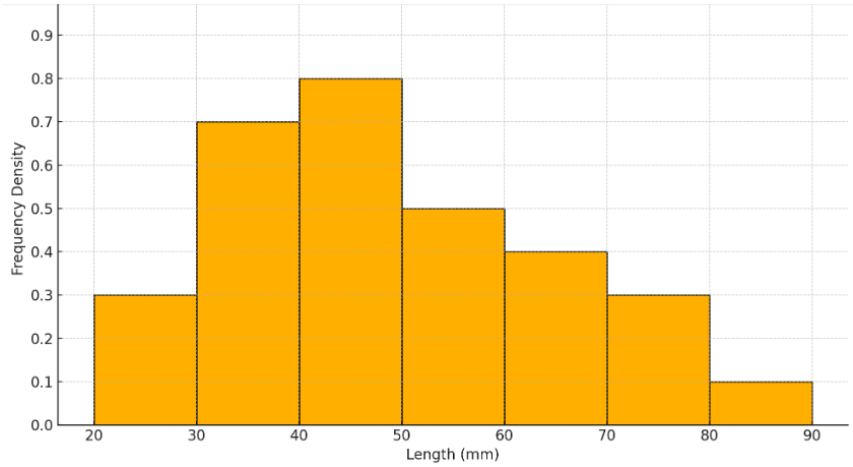
$$QS_k = \frac{(Q_3 - Q_2) - (Q_2 - Q_1)}{Q_3 - Q_1} = \frac{13 - 11}{61 - 37} = 0.083 > 0.$$

This indicates a positive skew. The frequency distribution is shown below, together with the histogram. Since each interval width is 10,

$$frequency\ density = frequency \div 10$$

Length (mm)	Frequency	Frequency density
$20 \leq x < 30$	3	0.3
$30 \leq x < 40$	7	0.7
$40 \leq x < 50$	8	0.8
$50 \leq x < 60$	5	0.5

$60 \leq x < 70$	4	0.4
$70 \leq x < 80$	3	0.3
$80 \leq x < 90$	1	0.1



This histogram also confirms the positive skew.

3. In a frequency distribution the co-efficient of skewness based upon the quartiles is 0·6. If the sum of the upper and lower quartiles is 100 and median is 38, find the value of the upper the upper and lower quartiles is 100 and median is 38, find the value of the upper and lower quartiles.

SOLUTION

$$QS_k = \frac{Q_3 + Q_1 - 2Q_2}{Q_3 - Q_1} = 0.6$$

Also Sum: $Q_3 + Q_1 = 100,\ median(Q_2) = 38$, substituting we get,

$$\Rightarrow \frac{100 - 2 \times 38}{Q_3 - Q_1} = 0.6$$

$$\Rightarrow \frac{24}{Q_3 - Q_1} = 0.6$$

$$\Rightarrow 0.6(Q_3 - Q_1) = 24$$

$$\Rightarrow Q_3 - Q_1 = \frac{24}{0.6} = 40$$

$\Rightarrow Q_3 = 40 + Q_1$, now substituting Q_3 into $Q_3 + Q_1 = 100$, gives

$$40 + Q_1 + Q_1 = 100$$

$$2Q_1 = 100 - 40 \Rightarrow Q_1 = \frac{60}{2} = 30 \therefore\ Q_3 = 70$$

1.4. MOMENTS

- ✓ The r moment of a varipblc x about any point $x = A$, usually denoted $\boldsymbol{\mu_r'}$ is given by

$$\mu_r' = \frac{1}{N}\sum_i f_i\,(x_i - A)^r, N = \sum_i f_i$$

$$= \frac{1}{N}\sum_i f_i\,d_i^r, \qquad \text{where } d_i = (x_i - A)$$

- ✓ The r moment of a varipblc about the mean $\bar{x}$, usually denoted $\boldsymbol{\mu_r}$ is given by

$$\mu_r = \frac{1}{N}\sum_i f_i\,(x_i - \bar{x})^r = \frac{1}{N}\sum_i f_i\,z_i^r, \qquad \text{where } z_i = (x_i - \bar{x})$$

- ✓ In particular

$$\mu_0 = \frac{1}{N}\sum_i f_i\,(x_i - \bar{x})^0 = \frac{1}{N}\sum_i f_i = 1$$

$$\mu_1 = \frac{1}{N}\sum_i f_i\,(x_i - \bar{x}) = 0,$$

(being the algebraic sum of deviations from the mean)

$$\mu_2 = \frac{1}{N}\sum_i f_i\,(x_i - \bar{x})^2 = \sigma^2(variance),$$

- ✓ These results: $\mu_0 = 1,\ \mu_1 = 0, \text{and } \mu_2 = \sigma^2$ are of fundamental importance and should be committed to memory.
- ✓ **NOTE**

If $d_i = (x_i - A)$, then

$$\bar{x} = A + \frac{1}{N}\sum_i f_i\,d_i = A + \mu_1'$$

1.5. RELATION BETWEEN"MOMENTS ABOUT MEAN IN TERMS OF MOMENTS ABOUT ANY POINT AND *VICE VERSA*.

- ✓ We have

$$\mu_r = \frac{1}{N}\sum_i f_i\,(x_i - \bar{x})^r = \mu_r = \frac{1}{N}\sum_i f_i\,(x_i - A + A - \bar{x})^r$$

$$= \frac{1}{N}\sum_i f_i\,(d_i + A - \bar{x})^r, \text{where } d_i = (x_i - A)$$

Since,

$$\bar{x} = A + \frac{1}{N}\sum_i f_i\,d_i = A + \mu_1' \Longrightarrow \bar{x} = A + \mu_1' \Longrightarrow \bar{x} - A = \mu_1'$$

Then we have that

$$\mu_r = \frac{1}{N}\sum_i f_i\,(d_i + A - \bar{x})^r = \frac{1}{N}\sum_i f_i\,(d_i - \mu_1')^r, \text{where } A - \bar{x} = -\mu_1'$$

$$= \frac{1}{N}\sum_i f_i\,[d_i^r - {}^rC_1 d_i^{r-1}\mu_1' + {}^rC_2 d_i^{r-2}\mu_1'^2 - {}^rC_3 d_i^{r-3}\mu_1'^3 + \cdots + (-1)^r\mu_1'^r\,]$$

When expanding μ_r and using the result of $\mu_r' = \frac{1}{N}\sum_i f_i\,(x_i - A)^r = \frac{1}{N}\sum_i f_i\,d_i^r$, we get: $\mu_r = \mu_r' - {}^rC_1\mu_{r-1}'\mu_1' + {}^rC_2\mu_{r-2}'\mu_1'^2 - \cdots + (-1)^r\mu_1'^r$

In particular, on putting $r = 2, 3$ and 4 in the eqn for $\boldsymbol{\mu_r}$, we get

$$\mu_2 = \mu_2' - \mu_1'^2$$
$$\mu_3 = \mu_3' - 3\mu_2'\mu_1' + 2\mu_1'^3$$
$$\mu_4 = \mu_4' - 4\mu_3'\mu_1' + 6\mu_2'\mu_1'^2 - 3\mu_1'^4$$

Conversely,

$$\mu_r' = \frac{1}{N}\sum_i f_i\,(x_i - A)^r = \mu_r = \frac{1}{N}\sum_i f_i\,(x_i - \bar{x} + \bar{x} - A)^r$$

$$= \frac{1}{N}\sum_i f_i\,(z_i + \mu_1')^r, \text{where } z_i = (x_i - \bar{x}) \text{ and } \bar{x} = A + \mu_1'$$

Thus,

$$\mu_r' = \frac{1}{N}\sum_i f_i\,[z_i^r + {}^rC_1 z_i^{r-1}\mu_1' + {}^rC_2 z_i^{r-2}\mu_1'^2 + {}^rC_3 z_i^{r-3}\mu_1'^3 + \cdots + \mu_1'^r\,]$$

When expanding $\boldsymbol{\mu_r'}$ and using the result of $\mu_r = \frac{1}{N}\sum_i f_i\,(x_i - \bar{x})^r = \frac{1}{N}\sum_i f_i\,z_i^r$, we get: $\mu_r' = \mu_r + {}^rC_1\mu_{r-1}\mu_1' + {}^rC_2\mu_{r-2}\mu_1'^2 + \cdots + \mu_1'^r$

In particular, putting $r = 2, 3$ and 4 and noting that $\mu_1 = 0$, we get

$$\mu_2' = \mu_2 + \mu_1'^2$$
$$\mu_3' = \mu_3 + 3\mu_2\mu_1' + \mu_1'^3$$
$$\mu_4' = \mu_4 + 4\mu_3\mu_1' + 6\mu_2\mu_1'^2 + \mu_1'^4$$

These formulae enable us to find the moments about any point, once the mean and the moments about mean are known.

1.6. PEARSON'S $\boldsymbol{\beta}$ AND $\boldsymbol{\gamma}$ COEFFICIENTS.

- ✓ Karl Pearson defined the following four coefficients, based upon the first four moments about mean:

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3}, \qquad \gamma_1 = +\sqrt{\beta_1}, \text{ and } \beta_2 = \frac{\mu_4}{\mu_2^2}, \qquad \gamma_2 = \beta_2 - 3$$

- ✓ It may be pointed out that these coefficients are pure numbers independent of units of measurement.

REMARK

Sometimes, another coefficient based on moments, namely, Alpha (α) coefficient is used. Alpha coefficients are defined as:

$$\alpha_1 = \frac{\mu_1}{\sigma} = 0,\ \alpha_2 = \frac{\mu_2}{\sigma^2} = 1,\ \alpha_3 = \frac{\mu_3}{\sigma^3} = \sqrt{\beta_1} = \gamma_1,\ \alpha_4 = \frac{\mu_4}{\sigma^4} = \beta_2$$

EXAMPLES

1. The first four moments of a distribution about the value 4 of the variable are $-1.5, 17, -30$ and, 108. Find the moments about mean, β_1 and β_2. Find also the moments about (i) the origin and (ii) the point $x = 2$.

   SOLUTION

   In the usual notations, we are given $A = 4$ and $\mu_1' = -1.5,\ \mu_2' = 17,$ $\mu_3' = -30,$ and $\mu_2' = 108.$

   Moments about mean: $\mu_1 = 0$

   $$\mu_2 = \mu_2' - \mu_1'^2 = 17 - (-1.5)^2 = \mathbf{14.75}$$

   $$\mu_3 = \mu_3' - 3\mu_2'\mu_1' + 2\mu_1'^3$$

   $$= -30 - 3(17)(-1.5) + 2(-1.5)^3 = \mathbf{39.75}$$

   $$\mu_4 = \mu_4' - 4\mu_3'\mu_1' + 6\mu_2'\mu_1'^2 - 3\mu_1'^4$$

   $$= 108 - 4(-30)(-1.5) + 6(17)(-1.5)^2 - 3(-1.5)^4$$

   $$= 108 - 180 + 229.5 - 15.1875 = \mathbf{142.3125}$$

   Here

   $$\beta_1 = \frac{\mu_3^2}{\mu_2^3} = \frac{(39.75)^2}{(14.75)^3} = 0.4924,$$

   $$\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{142.3125}{(14.75)^2} = 0.6541$$

   Also, $\bar{x} = A + \mu_1' = 4 + (-1.5) = 2.5$

   (i) Moments about origin. We have

   $\bar{x} = 2.5,\ \mu_2 = 14.75,\ \mu_3 = 39.75$ and $\mu_4 = 142.31(approx)$

   We know $\bar{x} = A + \mu_1'$ where μ_1' is the first moment about the point $x = A$.

   Taking $A = 0$, we get the first moment about origin as $\mu_1' = mean = 2 \cdot 5.$

   $$\mu_2' = \mu_2 + \mu_1'^2 = 14.75 + (2.5)^2 = 14.75 + 6.25 = 21$$

   $$\mu_3' = \mu_3 + 3\mu_2\mu_1' + \mu_1'^3 = 39.75 + 3(14.75)(2.5) + (2.5)^3$$

   $$= 39.75 + 110.625 + 15.625 = 166$$

   $$\mu_4' = \mu_4 + 4\mu_3\mu_1' + 6\mu_2\mu_1'^2 + \mu_1'^4$$

   $$= 142.3125 + 4(39.75)(2.5) + 6(14.75)(2.5)^2 + (2.5)^4$$

   $$= 142.3125 + 397.5 + 553.125 + 39.0625 = 1132$$

***(ii) Moments about the point x* = 2**. We have $\bar{x} = A + \mu_1'$. Taking $A = 2$, the first moment about the point $x = 2$ is: $\mu_1' = \bar{x} - A = 2.5 - 2 = 0.5$

$$\mu_2' = \mu_2 + \mu_1'^2 = 14.75 + (0.5)^2 = 14.75 + 0.25 = 15$$

$$\mu_3' = \mu_3 + 3\mu_2\mu_1' + \mu_1'^3 = 39.75 + 3(14.75)(0.5) + (0.5)^3$$

$$= 39.75 + 22.125 + 0.125 = 62$$

$$\mu_4' = \mu_4 + 4\mu_3\mu_1' + 6\mu_2\mu_1'^2 + \mu_1'^4$$

$$= 142.3125 + 4(39.75)(0.5) + 6(14.75)(0.5)^2 + (0.5)^4$$

$$= 142.3125 + 79.5 + 22.125 + 0.0625 = 244$$

2. Calculate the first four moments of the following distribution about the mean and hence find β_1 and β_2.

x	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
f	1	8	28	56	70	56	28	8	1

SOLUTION

x	f	$d = x - 4$	fd	fd^2	fd^3	fd^4
0	1	-4	-4	16	-64	256
1	8	-3	-24	72	-216	648
2	28	-2	-56	112	-224	448
3	56	-1	-56	56	-56	56
4	70	0	0	0	0	0
5	56	1	56	56	56	56
6	28	2	56	112	224	448
7	8	3	24	72	216	648
8	1	4	4	16	64	256
Total	$N = \sum f = 256$	0	0	512	0	2816

Moments about the points $x = 4$ are

$$\mu_1' = \frac{1}{N}\sum fd = 0,\ \mu_2' = \frac{1}{N}\sum fd^2 = \frac{512}{256} = 2,\ \mu_3' = \frac{1}{N}\sum fd^3 = 0 \text{ and}$$

$$\mu_4' = \frac{1}{N}\sum fd^4 = \frac{2816}{256} = 11.$$

Moments about mean are:

$$\mu_1 = 0$$

$$\mu_2 = \mu_2' - \mu_1'^2 = 2$$

$$\mu_3 = \mu_3' - 3\mu_2'\mu_1' + 2\mu_1'^3 = 0$$

$$\mu_4 = \mu_4' - 4\mu_3'\mu_1' + 6\mu_2'\mu_1'^2 - 3\mu_1'^4 = 11$$

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3} = 0$$

$$\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{11}{4} = 2.75$$

1.7. COEFFICIENT OF SKEWNESS BASED ON MOMENTS

- ✓ The coefficient of skewness based on moment is given by

$$S_k = \frac{\sqrt{\beta_1}(\beta_2 + 3)}{2(5\beta_2 - 6\beta_1 - 9)}$$

- ✓ Where, Karl Pearson defined the following four coefficients, based upon the first four moments about mean as follows:

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3}, \qquad \gamma_1 = +\sqrt{\beta_1}, \text{ and } \beta_2 = \frac{\mu_4}{\mu_2^2}, \qquad \gamma_2 = \beta_2 - 3$$

- ✓ Once again, it may be pointed out that these coefficients are pure numbers independent of units of measurement.

NOTE

1. $s_k = 0$, if either $\beta_1 = 0$ *or* $\beta_2 = -3$. But since $\beta_2 = \frac{\mu_4}{\mu_2^2}$, cannot be negative, $s_k = 0$ if and only if $\beta_1 = 0$. Thus, for a symmetrical distribution $\beta_1 = 0$. In this respect β_1 is taken to be a measure of skewness.
2. We observe that if $s_k > 0$, then the distribution is positively skewed.
3. Also, if $s_k < 0$, then the distribution is negatively skewed.

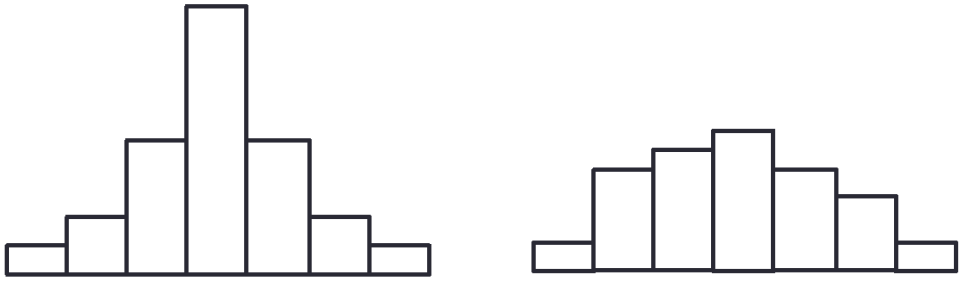
2. KURTOSIS

- ✓ **Kurtosis** or **convexity of curve** is the measure of the peakedness of a frequency distribution relative to the normal distribution.
- ✓ In other words, kurtosis enables us to have an idea about the flatness or peakedness of the curve or distribution.
- ✓ It is measured by the co-efficient β_2 given by

$$\beta_2 = \frac{\mu_4}{\mu_2^2}, \qquad \gamma_2 = \beta_2 - 3.$$

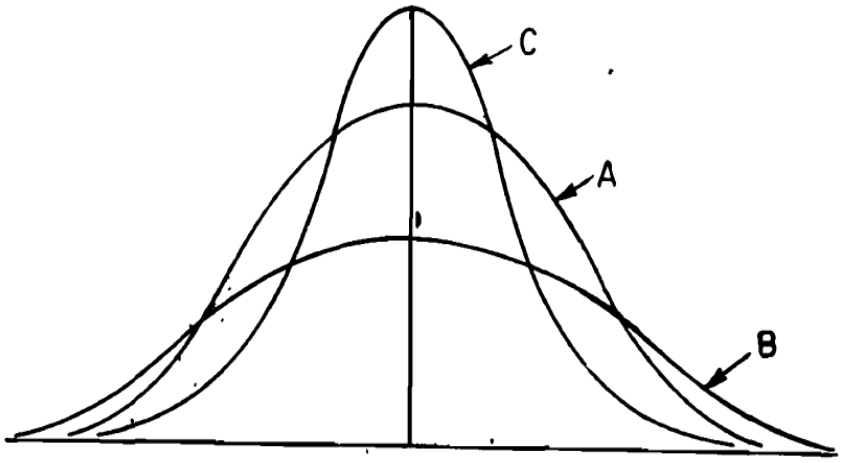
- ✓ It is important to note that the larger the kurtosis, the more peaked will be the distribution.
- ✓ There are two types of kurtoses under discussion, these are, **Leptokurtic** and **platykurtic.**
- ✓ **A Leptokurtic distribution:** This is a more peaked distribution than a normal distribution**.**

- ✓ **A platykurtic distribution:** This is described as a flatter distribution compared to the normal distribution**.**
- ✓ The following graphs below show both the leptokurtic and platykurtic types of kurtoses.



(a) Leptokurtic distribution (b) Platykurtic distribution

- ✓ We provide the summary of kurtosis by discussing the diagram below.



1. Curve of the type *'A'* which is neither flat nor peaked is called the ***normal curve or mesokurtic curve*** and for such a curve $\beta_2 = 3,\ i.e.,\ \gamma_2 = 0.$
2. Curve of the type *'B'* which is flatter than the normal curve is known as ***platykurtic*** and for such a curve $\beta_2 < 3,\ i.e.,\ \gamma_2 < 0.$
3. Curve of the type 'C' which is more peaked than the normal curve is called ***leptokurtic*** and for such a curve $\beta_2 > 3,\ i.e.,\ \gamma_2 > 0.$

EXAMPLE

For a distribution the mean is 10, variance is 16, $\gamma_2 = +1$ is β_2 is 4. (i) Obtain the first four moments about the origin, i.e., zero. (ii) Comment upon the nature of distribution.

SOLUTION

We are given

$mean = 10, \mu_2 = 16,\ \beta_2 = 4, \gamma_1 = +1$

(i) First four moments about origin $(\mu_1', \mu_2', \mu_3', \mu_4')$

μ_1' = First moment about origin $= Mean = 10$

$\mu_2 = \mu_2' - \mu_1'^2 \Longrightarrow \mu_2' = \mu_2 + \mu_1'^2 \Longrightarrow \mu_2' = 16 + 10^2 = 116$`

Since

$$\gamma_1 = +\sqrt{\beta_1} \text{ and } \beta_1 = \frac{\mu_3^2}{\mu_2^3} \Longrightarrow \gamma_1 = +1 \Longrightarrow \frac{\mu_3}{\mu_2^{3/2}} = 1$$

$$\Longrightarrow \mu_3 = \mu_2^{3/2} = (16)^{3/2} = 4^3 = 64$$

$$\therefore \mu_3 = \mu_3' - 3\mu_2'\mu_1' + 2\mu_1'^3$$

$$\Longrightarrow \mu_3' = \mu_3 + 3\mu_2'\mu_1' - 2\mu_1'^3 = 64 + 3(116)(10) - 2(10)^3$$

$$= 3544 - 2000 = 1544$$

Now,

$$\beta_2 = \frac{\mu_4}{\mu_2^2} = 4 \implies \frac{\mu_4}{16^2} = 4 \Longrightarrow \mu_4 = 4 \times 16^2 = 1024$$

And

$$\mu_4 = \mu_4' - 4\mu_3'\mu_1' + 6\mu_2'\mu_1'^2 - 3\mu_1'^4$$

$$\Longrightarrow \mu_4' = \mu_4 + 4\mu_3'\mu_1' - 6\mu_2'\mu_1'^2 + 3\mu_1'^4$$

$$= 1024 + 4(1544)(10) - 6(116)(100) + 3(10000)$$

$$= 92784 - 69600 = 23184$$

(ii) Comments on nature of distribution

Since $\gamma_1 = +1$ the distribution is moderately positively skewed, i.e., if we draw the curve for the given distribution, it will have longer tail towards the right. Further since $\beta_2 = 4 > 3$, the distribution is leptokurtic, i.e., it will be more peaked than the normal curve.

✓ **TASK**

1. What do you understand by skewness? How is it measured? Distinguish clearly, by giving figures, between positive and negative skewness
2. Explain the methods of measuring skewness and kurtosis of a frequency distribution.
3. Show that for any frequency distribution:
   (i) kurtosis is *greater* than unity
   (ii) Co-efficient of skewness is *less* than 1 numerically.

3. CURVE FITTING

✓ Curve fitting is the process of finding a mathematical function that best fits a series of data points.

✓ The goal of curve fitting is to find a curve (like a line, polynomial, exponential, etc.) that captures the underlying pattern or trend of the data.

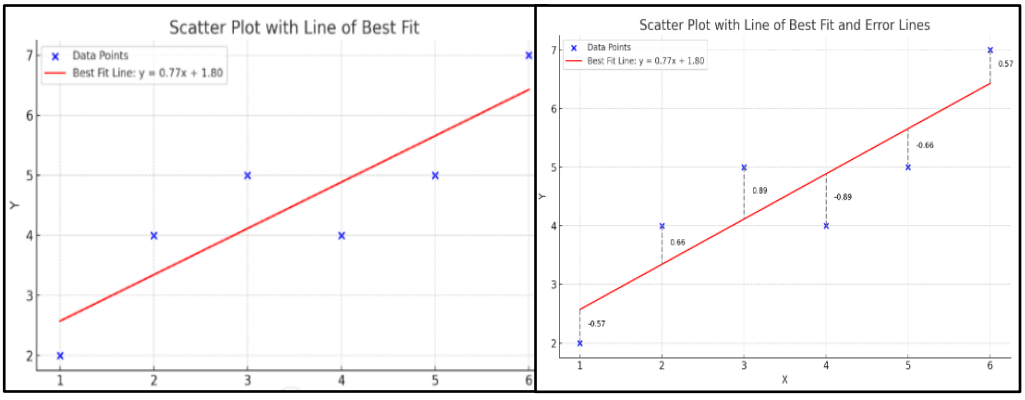
- ✓ The general problem in curve fitting is to find, if possible, an analytic expression of the form $y = f(x)$, for the functional relationship suggested by the given data.

3.1. WHY IS CURVE FITTING IMPORTANT?

- ✓ Fitting of curves to a set of numerical data is of considerable importance for both theoretical as well as practical.
- ✓ Theoretically it is useful in the study of correlation and regression, *e.g.*, lines of regression can be regarded as fitting of linear curves to the given bivariate distribution.
- ✓ In practical statistics it enables us to represent the relationship between two variables by simple algebraic expressions, *e.g.*, polynomials, exponential or logarithmic functions.
- ✓ Moreover, it may be used to estimate the values of one variable which would correspond to the specified values of the other variable.
- ✓ The following are the summary reasons for the importance of curve fitting:
  (i)To make predictions (extrapolate or interpolate).
  (ii)To understand relationships between variables.
  (iii)To simplify complex data into an analytical expression.

3.2. FITTING OF A STRAIGHT LINE

- ✓ Let us consider the fitting of a straight line $Y = a + bX$ to a set of n points (x_i, y_i); $i = 1,2,\ldots,n$, where a and b are arbitrary constants.
- ✓ The problem here is to determine $'a'$ and $'b'$ so that a straight line $Y = a + bX$ becomes the line of **best fit**.
- ✓ The term $'$**best fit**$'$ is interpreted in accordance with the principle of least squares which consists of minimising the sum of the squares of the deviations of the actual values of y from their estimated values as given by the line of **best fit**.
- ✓ In other words, the principle of least squares method is a mathematical approach to determine the best-fitting curve by minimizing the sum of the squares of the vertical distances (errors) between the observed values and the values predicted by the model as shown in the scatter diagram below:



- ✓ Where

  (i) Blue dots are the actual observed data points

  (ii) Red line is the line of best fit, calculated using the Least Squares Method.

  (iii) Gray dashed lines are vertical distances (errors or residuals) between each observed point and the line of best fit.

  (iv) Text labels are the numerical values of each residual (i.e., how far off the prediction was from the actual value).

- ✓ Each vertical gray line represents the error for one point:

$$Error_i = y_i - \hat{y}_i$$

- ✓ Where:

  y_i is the actual data value.

  $\hat{y}_i = a + bx_i$ is the predicted value from the line of best fit.

- ✓ The **least squares method** minimizes the **sum of the squares of these errors** to ensure the best-fitting line.

  NOTE

  The least squares principle approach is used to find $'a'$ and $'b'$ such that sum of the squares of the vertical distances (errors) or sum of the squares of the deviations is minimized. This is typically done using calculus or matrix algebra.

- ✓ Thus, according to the principle of least squares, we have to determine a and b so that

$$E = \sum_{i=1}^{n} (y_i - a - bx_i)^2$$

  is minimum.

- ✓ From the principle of maxima and minima, the partial derivatives of *E*, with respect to (w.r.t.) a and b should vanish separately, *i.e.*,

$$\frac{\partial E}{\partial a} = 0 = -2\sum_{i=1}^{n}(y_i - a - bx_i) \text{ and } \frac{\partial E}{\partial b} = 0 = -2\sum_{i=1}^{n} x_i(y_i - a - bx_i)$$

$$\Rightarrow \sum y_i = na + b\sum x_i \text{ and } \sum x_i y_i = a\sum x_i + b\sum x_i^2$$

- ✓ The two Equations are known as the **normal equations** for estimating a and b.
- ✓ All quantities $\sum x_i$, $\sum x_i^2$, $\sum y_i$ and $\sum x_i y_i$ can be obtained by the set of points (x_i, y_i); $i = 1,2,\ldots,n$, and the above normal equations can be solved for a and b.
- ✓ With the values of *a* and *b* so obtained, equation $Y = a + bX$ is the line of best fit to the given set of points (x_i, y_i); $i = 1,2,\ldots,n$.

NOTE

The equation of the line of best fit of Y on X is obtained on eliminating a and b in the normal equations above and can be expressed in the determinant form as follows:

$$\begin{vmatrix} Y & X & 1 \\ \sum y_i & \sum x_i & n \\ \sum x_i y_i & \sum {x_i}^2 & \sum x_i \end{vmatrix} = 0$$

EXAMPLE

Fit a straight line to the following data.

X	1	2	3	4	6	8
Y	2.4	3	3.6	4	5	6

SOLUTION

Let the line be $Y = a + bX$

X	Y	X^2	XY
1	2.4	1	2.4
2	3.0	4	6.0
3	3.6	9	10.8
4	4.0	16	16.0
6	5.0	36	30.0
8	6.0	64	48.0
$\sum X = 24$	$\sum Y = 24$	$\sum X^2 = 130$	$\sum XY = 113.2$

Using the normal equations:

$$\sum y_i = na + b\sum x_i \text{ and } \sum x_i y_i = a\sum x_i + b\sum x_i^2$$

we get

$24 = 6a + 24b$ and $113.2 = 24a + 130b$

Solving these equations, we get $a = 1.976$ and $b = 0.506$.

Thus, the required equation of a straight line is

$$Y = 1.976 + 0.506X$$

3.3. FITTING OF SECOND-DEGREE PARABOLA

- ✓ Let $Y = a + bX + cX^2$ be the second-degree parabola of best fit to set of n points $(x_i, y_i);\ i = 1,2,\ldots,n.$
- ✓ Using the principle of least squares, we have to determine the constants a, b and c, so that

$$E = \sum_{i=1}^{n} \left(y_i - a - bx_i - cx_i^2\right)^2$$

is minimum.

- ✓ Thus, equating to zero the partial derivatives of E with respect to $a,\ b$ and c separately, we get the normal equations for estimating a, b and c are obtained as

$$\frac{\partial E}{\partial a} = 0 = -2\sum_{i=1}^{n}\left(y_i - a - bx_i - cx_i^2\right)$$

$$\frac{\partial E}{\partial b} = 0 = -2\sum_{i=1}^{n} x_i\left(y_i - a - bx_i - cx_i^2\right)$$

$$\frac{\partial E}{\partial a_k} = 0 = -2\sum_{i=1}^{n} x_i^2\left(y_i - a - bx_i - cx_i^2\right)$$

$$\Longrightarrow$$

$$\sum y_i = na + b\sum x_i + c\sum x_i^2$$

$$\sum x_i y_i = a\sum x_i + b\sum x_i^2 + c\sum x_i^3$$

$$\sum x_i^2 y_i = a\sum x_i^2 + b\sum x_i^3 + c\sum x_i^4$$

- ✓ summation taken over i from $1\ to\ n$.
- ✓ For given sets of points $(x_i, y_i);\ i = 1, 2, \ldots, n$, the above equations can be solved for a, b and c, and with these values of a, b and c, is the parabola of best fit.

NOTE

Eliminating a, b and c in the above equations, the parabola of best fit of Y on X is given by

$$\begin{vmatrix} Y & X^2 & X & 1 \\ \sum y_i & \sum x_i{}^2 & \sum x_i & n \\ \sum x_i y_i & \sum x_i{}^3 & \sum x_i^2 & \sum x_i \\ \sum x_i^2 y_i & \sum x_i{}^4 & \sum x_i^3 & \sum x_i{}^2 \end{vmatrix} = 0$$

EXAMPLE

Fit a parabola of second degree to the following data.

X	0	1	2	3	4
Y	1	1.8	1.3	2.5	6.3

SOLUTION

Let $Y = a + bX + cX^2$ be the second-degree parabola.

	X	Y	X^2	X^3	X^4	XY	X^2Y
	0	1.0	0	0	0	0	0
	1	1.8	1	1	1	1.8	1.8

	2	1.3	4	8	16	2.6	5.2
	3	2.5	9	27	81	7.5	22.5
	4	6.3	16	64	256	25.2	100.8
Total	10	12.9	30	100	354	37.1	130.3

Using the normal equations:

$\sum y_i = na + b\sum x_i + c\sum x_i^2$

$\sum x_i y_i = a\sum x_i + b\sum x_i^2 + c\sum x_i^3$

$\sum x_i^2 y_i = a\sum x_i^2 + b\sum x_i^3 + c\sum x_i^4$

we get

$12 \cdot 9 = 5a + l0b + 30c;\ 37 \cdot 1 = 10a + 30b + 100c;$

$130 \cdot 3 = 30a + 100b + 354c$

Solving these equations, we get $a = 1 \cdot 42, b = -1 \cdot 07$ and $c = 0.55$. Thus the required equation of the second-degree parabola is

$$Y = 1 \cdot 42 - 1 \cdot 07X + 0.55X^2$$

3.4. FITTING OF POLYNOMIAL OF Kth DEGREE.

- ✓ If $Y = a_0 + a_1X + a_2X^2 + \cdots + a_kX^k$ is the kth degree polynomial of best fit to the set of points $(x_i, y_i);\quad i = 1, 2, \ldots, n$, the constants $a_0, a_1, \ldots, a_k$ are to be obtained so that

$$E = \sum_{i=1}^{n} \left(y_i - a_0 - a_1x_i - a_2x_i^2 - \cdots - a_kx_i^k\right)^2$$

  is minimum.

- ✓ Thus, the normal equations for estimating $a_0, a_1, \ldots, a_k$ are obtained by equating to zero the partial derivatives of E w.r.t. $a_0, a_1, \ldots, a_k$ separately, i.e.,

$$\frac{\partial E}{\partial a_0} = 0 = -2\sum_{i=1}^{n}\left(y_i - a_0 - a_1x_i - a_2x_i^2 - \cdots - a_kx_i^k\right)$$

$$\frac{\partial E}{\partial a_1} = 0 = -2\sum_{i=1}^{n} x_i\left(y_i - a_0 - a_1x_i - a_2x_i^2 - \cdots - a_kx_i^k\right)$$

$$\frac{\partial E}{\partial a_k} = 0 = -2\sum_{i=1}^{n} x_i^k\left(y_i - a_0 - a_1x_i - a_2x_i^2 - \cdots - a_kx_i^k\right)$$

$\Rightarrow$

$\sum y_i = na_0 + a_1\sum x_i + a_2\sum x_i^2 + \cdots + a_k\sum x_i^k$

$\sum x_i y_i = a_0\sum x_i + a_1\sum x_i^2 + a_2\sum x_i^3 + \cdots + a_k\sum x_i^{k+1}$

$\sum x_i^k y_i = a_0\sum x_i^k + a_1\sum x_i^{k+1} + a_2\sum x_i^{k+2} + \cdots + a_k\sum x_i^{2k}$

summation extended over i from 1 $to\ n$. These are $(k\ +\ 1)$ equations in $(k\ +\ 1)$ unknowns $a_0, a_1, \dots, a_k$ and can be solved with the help of algebra.

NOTE

It has been found that in all the above cases, the values of the second order derivatives,

$$\frac{\partial^2 E}{\partial a_0^2}, \frac{\partial^2 E}{\partial a_1^2}, \dots$$

come out to be positive at the points $a_0, a_1, \dots, a_k$ the solutions of the 'normal equations'. Hence, they provide minima of *E*.

3.5. APPLICATIONS OF CURVE FITTING & LEAST SQUARES

(i) Weather and climate modelling
(ii) Economics and finance
(iii) Machine learning regression models
(iv) Physics and engineering experiments
(v) Epidemiology and population studies

SUMMARY

Concept	Description
Curve Fitting	Finding a mathematical function that best fits a dataset
Least Squares	A method to minimize the sum of squared errors (residuals)
Error (Residual)	Difference between observed value and model prediction
Best Fit	Achieved when the total squared error is minimized

3.6. GENERAL CURVE FITTING (NOT JUST LINEAR)

✓ Curve fitting using least squares is not limited to straight lines. You can fit:

(i) Polynomials: $y = a_n x^n + \cdots + a_1 x + a_0$
(ii) Exponentials: $y = ae^{bx}$
(iii) Logarithmic: $y = alog(x) + b$
(iv) Power Laws, Sine Functions, etc.